MIL-F-51318A(EA)
29 March 1988
SUPERSEDING
MIL-F-51318(MU)
6 November 1968

MILITARY SPECIFICATION

FILTER UNIT, GAS-PARTICULATE, EMD, 300 CFM, M6A2

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electric motor driven (EMD) filter unit.

2* APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed In the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation-

SPECIFICATIONS

FEDERAL

P-D-680	- Dry Cleaning Solvent.
FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
TT-W-572	- Wood-Presenative: Water-Repellent.
PPP-B-636	- Boxes, Shipping, Fiberboard.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock.
PPP-T-97 - Tape, Packaging/Industrial, Filament Reinforced.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-G-17504 - Coating Compound, Acrylic, Clean.
MIL-T-22085 - Tapes, Pressure-Sensitive, Adhesive, Preservation.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load Bearing Base.
MIL-R-46089 - Rubber, Sponge, Silicone, Closed Cell.
MIL-A-46146 - Adhesive Sealants, Silicone, RTV, Noncorrosive.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-19-5379 - Filter Unit, Gas-Particulate, EMD, 300 CFM, M6A2.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-673 - Filter, Gas.
P5-19-1821 - Filter, Particulate.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting. activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-5379 or on the subsidiary drawings shall conform to the specifications listed therein or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the filter unit shall conform to the specification and drawings listed on Drawing 5-19-5379 and subsidiary drawings.

3.1.2.1 Electric motor driven blower. A performance curve of head pressure versus air flow, from the maximum air flow at 0 head down to 100 cfm, shall be provided with each electric rotor driven blower.

3.2 Manufacture and assembly. The filter unit shall be manufactured and assembled in accordance with Drawing 5-19-5379.

3.3 Impeller alignment. When the unit is operated for 1 minute connected for blow-through and 1 minute connected for draw-through, there shall be no evidence that the impeller is touching the impeller housing, when tested as specified in 4.2.4.1.

3.4 Workmanship. The filter unit shall be free from damage such as dents or cracks, end foreign matter such as dirt, oil, or grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performame of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any Inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contracts Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the M6A2 filter units produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of gas filters shall be represented in any one lot of finished filter units.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedures.

4.2.3.1 For examination and nondestructive tests. Sample packaged filter units shall be examined and tested in accordance with MIL-STD-105 and with the classification of defects below. Filter units shall be tested for performance (4.2.4) prior to packaging. Packaging inspection (4.2.3.2(c) shall be conducted prior to the examination of the filter units.

4.2.3.2 Classification of defects.

(a) Filter unit, gas-particulate, EMD, 300 CFM, M6A2 (Drawing 5-19-5379).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL 0.65 percent defective		
101		Impeller alignment	4.2.4.1
	AQL 1.0 percent defective		
102		Component missing, incorrect or incorrectly assembled	
103		Workmanship (see 3.4)	
Minor:	AQL 2.5 percent defective		
201		Protective finish missing or incorrect	
202		Marking missing, incorrect or illegible	
203		Electric motor driven blower performance curve missing	

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(b) Stand, engine (Drawing 5-19-1806).

Categories	Defects
Critical:	None defined
Major:	AQL 0.25 percent defective
101	Motor mounting surface perpendicular to surface A
102	Motor mounting surface parallel to surface B

(c) packging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Item not completely cleaned/dried
102	Painted and unpainted metal surfaces not treated as specified
103	Name and instruction plates and tags not coated as specified
104	Filter unit and blower assembly openings not covered as specified
105	Preservatives “not applied as specified
106	Filter unit components not packaged as specified or missing
107	Skid base not as specified
108	Strapping loose or missing
109	Marking missing, incorrect, or illegible
110	Waterproof shroud not as specified
111	Unit packs (filters) not packed as specified

4.2.4 Test.

4.2.4.1 Impeller alignment. The filter unit shall be connected as specified and operated for the required period Of ti-

5. PACKAGING

5.1 Cleaning and drying. The filter unit, including component arts thereto, shall be cleaned and dried in accordance with MIL-P-116, using clean ing method C-1 and any of the drying methods listed therein.

5.2 Preservation and unit packing, level A. All unit packs shall be of the minimum cube required to contain the items. Unless otherwise specified, preservations including the application thereof, shall be accomplished in accordance with the requirements of MIL-P-116.

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5.2.1 Painted and unpainted metal surfaces. Damaged portions of painted surfaces shall be treated and repainted In accordance with type A of MIL-T-704, using the same color or paint as specified on the drawing. Unpainted exterior noncritical and nonfunctioning surfaces, including threaded areas, shall be coated with type P-2 preservative.

5.2.1.1 Metal name and instruction plates, tags, and tag waterproofing. These items shall be coated with clear acrylic coating compound, conforming to MIL-C-17504. Cloth or paper tags shall conform to the applicable requirements of MIL-STD-129. The above specified coating shall not be applied to the tags until after printing.

5.2.1.2 Tape application. Painted surfaces to which pressure-sensitive tape is to be applied shall be cleaned with dry cleaning solvent conforming to type I of P-D-680. Unless otherwise specified herein, pressure-sensitive tape specified herein shall conform to type II of MIL-T-22085. Pressure-sensitive tape applied to painted surfaces shall be coated with acrylic coating compound conforming to MIL-C-17504.

5.2.1.3 Filter unit (less filters) and blower assembly. Small openings into the filter unit and blower assembly shall be sealed off with pressure-sensitive tape. Large openings lover two inches in diameter) shall be covered with a metal plate disk, sealing same in place with pressure-sensitive tape. The carrier handles shall be fully retracted and secured in this position with minimum 1-1/2 inch wide tape, conforming to type IV of PPP-T-97 and this tape shall be overcoated.

5.2.1.4 Electric motor. The electric motor shall be lubricated in accordance with the manufacturer's instructions. The shaft outside the motor and the rigid couplings shall be coated with preservative conforming to type P-2 of MIL-P-116. All openings into the electric motor shall be sealed off with pressure-sensitive tape.

5.2.1.5 Floor mat. The floor mat shall be rolled and tied with a suitable cord and put in a style RSC fiberboard box conforming to grade W6c or W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

5.2.1.6 Hose assembly with hose clamp. The hose shall be collapsed and placed on its carrier which is provided for this purpose. The hose and hose clamp shall be packaged in a style RSC fiberboard box conforming to grade W6c or W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

5.2.1.7 Air filter. The filter shall be packaged in a style RSC fiberboard box conforming to grade W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

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5.2.1.8 Mounting straps. The straps for mounting the air filter shall be packaged in a style RSC fiberboard box conforming to grade W6c or W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

5.2.1.9 Connector, washers, screws, and gaskets. The bolts, screws, and washers shall be packaged in accordance with method IC-1 of MIL-P-116. The connector, gasket, and the packaged bolts, screws, and washers shall be packaged in a style RSC fiberboard box conforming to grade W6c or W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

5.2.1.10 Technical publications. The technical publications shall be packaged in accordance with method IC-3 of MIL-P-116.

5.2.1.11 Filter, gas (Drawing 5-19-673), and filter, particulate (Drawing 5-19-1821). These filters shall be individually unit packed as specified in SPIs P5-19-673 and P5-19-1821. Two wood fillers, fabricated in accordance with Figure 1 on page 10 shall be secured in place with fasteners provided (one wood filler will replace two filters).

5.2.1.12 Consolidated packaging. Items packaged as specified in 5.2.1.5 through 5.2.1.10 shall be further packaged in a style RSC fiberboard box conforming to grade V3c of PPP-B-636. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636.

5.3 Packing. Levels to be as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Filter unit (less filters) (Drawing 5-19-5379). The filter unit including components, accessories, tools, and technical publications, preserved and unit packed as specified in 5.2 shall be packed in a box conforming to type II, style A, class 1 (full panel) of MIL-B-26195. Internal blocking, bracing, and strapping shall be accomplished in accordance with MIL-STD-1186, to prevent movement of the item within the shipping container. The consolidated package containing components, repair parts, accessories tools, and technical publications, shall be secured to the skid base of the crate with galvanized strapping, conforming to type I, class B of QQ-S-781 or secured to the equipment with tape in a location that will not cause damage to any of the mounted components on the filter unit. Tape used to secure the consolidated package to the equipment shall be a minimum of 1-1/2 inches wide, conforming to type IV of PPP-T-97. The tape shall not be overcoated. Prior to packing the filter unit within the box, a waterproof shroud fabricated from waterproofed Kraft mapping paper shroud material conforming to PPP-B-1055, class c-2(a), shall be placed over the filter unit. This waterproof shroud shall extend down all sides and ends of the unit to not less than 3 inches from the bottom of the unit.

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5.3.1.2 Filter, gas (Drawing 5-19-673) and filter particulate (Drawing 5-19-1821). Two each of these filters (see 6.3), unit packed as specified in 5.2.1.11, shall be packed for shipment in a skidded wood box conforming to class 2 (overseas), style 4, type 1 (easy load), 100-250 pound weight of contents of PPP-B-621. The unit packed filters shall be arranged in the shipping container on end. Fiberboard filler pads conforming to type CF, class weather-resistant, grade W5c of PPP-F-320 shall be included as necessary to make a tight pack. The box shall be closed and strapped in accordance with Appendix to PPP-B-621. Strapping and seals shall conform to QQ-S-781, strapping - type I, finish B, grade 2; seals - type D, style I or IV, zinc coated, class R, grade 2.

5.3.2 Level B.

5.3.2.1 Filter unit (less filters) (Drawing 5-19-5379). The filter unit, cleaned and dried as specified in 5.1, together with items preserved and unit packed as specified in 5.2 (less filters, 5.2.1.11) shall be packed as specified in 5.3.1.1, except lumber shall be grade domestic.

5.3.2.2 Filter, gas (Drawing 5-19-673) and filter, particulate (Drawing 5-19-1821). Two each of these filters, unit packed as specified in 5.2.1.10, shall be packed for shipment as specified in 5.3.1.2, except lumber shall be grade domestic.

5.4 Repair parts. When the filters (Drawings 5-19-673 and 5-19-1821) are procured for storage and issue as a repair part, unit packing, packing and marking shall be as specified on applicable SPI P5-19-673 or P5-19-1821. Additionally, when components of the filter unit are procured for storage and issue as repair parts, preservation, unit packing, packing and marking shall be as specified on their applicable SPI, which are identified by their part number preceded by “P”.

5.5 Marking. In addition to any special marking required by the contract or order, unit packs and packs of the filters (see 5.2 and 5.3) shall be marked in accordance with SPI P5-19-673 or SPI P5-19-1821, as applicable. Unit packs and packs of the filter unit (see 5.2 and 5.3) shall be marked in accordance with MIL-STD-129.

5.5.1 Special marking. Level A and B packs shall also be marked with the following information:

(a) Set or assembly markings.

(b) packing list (single stock number assembly).

(c) Shelf life markings.

(d) Packed date.

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6. NOTES

6.1 Intended use. The filter unit covered by this specification provides filtered air for Nike-Hercules installations and is used in Collective Protection Equipment, CBR; Nike-Hercules CONUS: Mll, M12, and M13.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.3).
(c) Shelf life code (see 5.5.1).

6.3 Component specifications. Component specifications for the M6A2 filter unit are:

Title	Number
Filter, Particulate, 150 CFM, M9A1	MIL-F-51165(EA)
Filter, Gas, 150 CFM, M10	MIL-F-0011137(EA)

6.4 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Blower, electric, motor driven
Collective protection equipment
Filter unit, gas-particulate

Custodian:

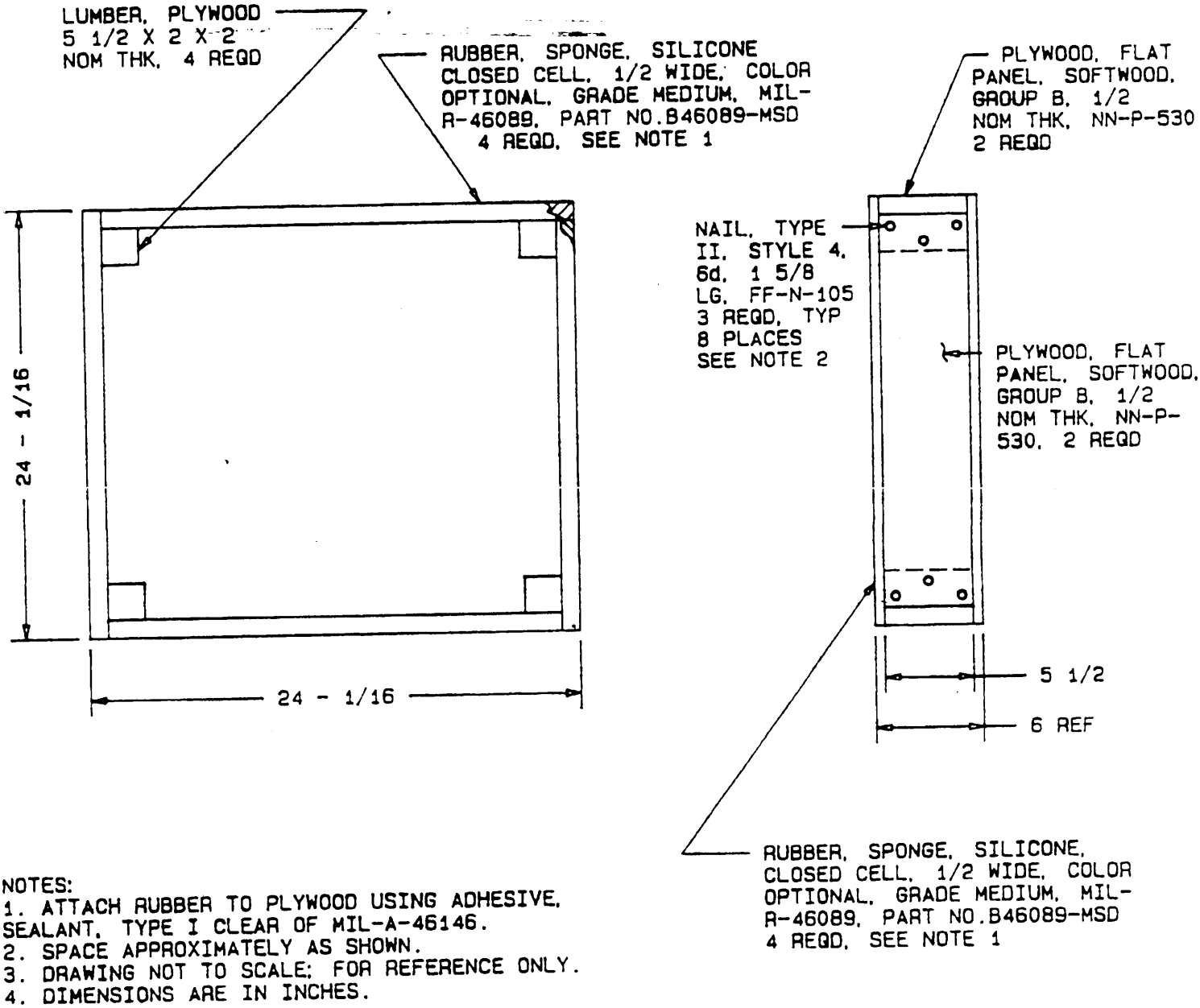
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NOTES:
1. ATTACH RUBBER TO PLYWOOD USING ADHESIVE, SEALANT, TYPE I CLEAR OF MIL-A-46146.
2. SPACE APPROXIMATELY AS SHOWN.
3. DRAWING NOT TO SCALE; FOR REFERENCE ONLY.
4. DIMENSIONS ARE IN INCHES.

FIGURE 1. Filler

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51318A(EA)	FILTER UNIT, GAS-PARTICULATE, EMD, 300 CFM, M6A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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